

GlaucoScan — RE-GAN Analysis Report

Fundus Image Preprocessing & Clinical Feature Extraction

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16.2%

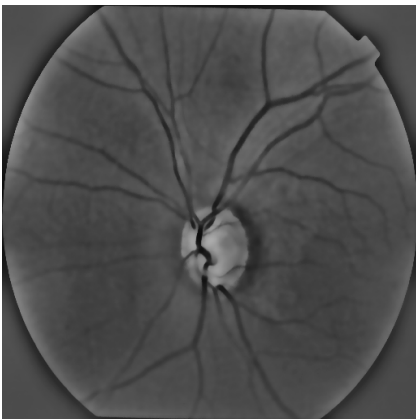
Low Risk Optic nerve head appears within normal limits. Routine annual eye examination recommended.

Preprocessing Results

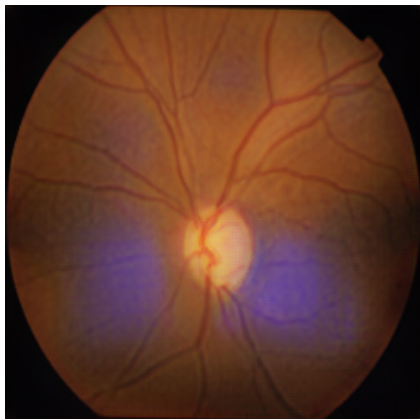
The fundus image was processed through a 5-stage classical pipeline followed by RE-GAN deep learning enhancement. The three images below show the progression from raw input to fully enhanced output.



Original (Raw input)



Classical Pipeline (Stages 1–5)



RE-GAN Enhanced (Deep learning)

Classical pipeline: 30.0 ms

RE-GAN enhancement: 2442.1 ms

Total: 2472.1 ms

Extracted Clinical Features

The following five biomarkers were extracted from the RE-GAN-enhanced image using the two-stage optic disc localisation algorithm. All features are grounded in published ophthalmological literature.

Feature	Value	Clinical Range	Interpretation	Weight
Vertical CDR	0.336	< 0.65 normal	Normal	50%
Horizontal CDR	0.483	< 0.65 normal	Borderline	—
Rim-Disc Area Ratio	0.818	> 0.50 normal	Normal rim tissue	25%
ISNT Score	2 / 3	3 = fully normal	Minor violation	15%
Vessel Density	0.33%	> 10% normal	Reduced vessel density	10%

Optic Disc & Cup Measurements

Measurement	Value	Measurement	Value
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Disc Height (px)	152.0	Cup Height (px)	51.1
Disc Width (px)	86.6	Cup Width (px)	41.8
Disc Centre X	137	Image Sharpness	18.47
Disc Centre Y	105	Enhancement	RE-GAN (U-Net+CBAM)

Glaucoma Risk Score Breakdown

The overall glaucoma probability of 16.2% is computed using a weighted clinical scoring formula based on published ophthalmological thresholds. Each feature contributes independently to the final score.

Feature	Raw Score	Weight	Contribution
Vertical CDR	5%	50%	2.5%
Rim-Disc Area Ratio	5%	25%	1.2%
ISNT Score	33%	15%	5.0%
Vessel Density	75%	10%	7.5%
TOTAL		100%	16.2%

Preprocessing Pipeline Used

Stage	Operation	Purpose	Status
1	Green Channel Extraction	Maximise disc/vessel contrast	✓ Applied
2	Gaussian BG Normalisation	Remove non-uniform illumination (Novel)	✓ Applied
3	CLAHE Enhancement	Boost local contrast	✓ Applied
4	Bilateral Denoising	Edge-preserving noise removal	✓ Applied
5	Gamma Correction	Brightness adjustment	✓ Applied
6	RE-GAN Enhancement	Deep learning enhancement (U-Net+CBAM)	✓ Applied

Important Notice

This report is generated by an automated research system (GlaucoScan RE-GAN) developed as part of a PhD research programme. It is intended for research and educational purposes only and does NOT constitute a medical diagnosis. All findings must be confirmed by a qualified ophthalmologist before any clinical decision is made. If you are experiencing vision problems, please consult a healthcare professional immediately.